

جامعة أم القرى
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Vertebral Column (The Low Back)

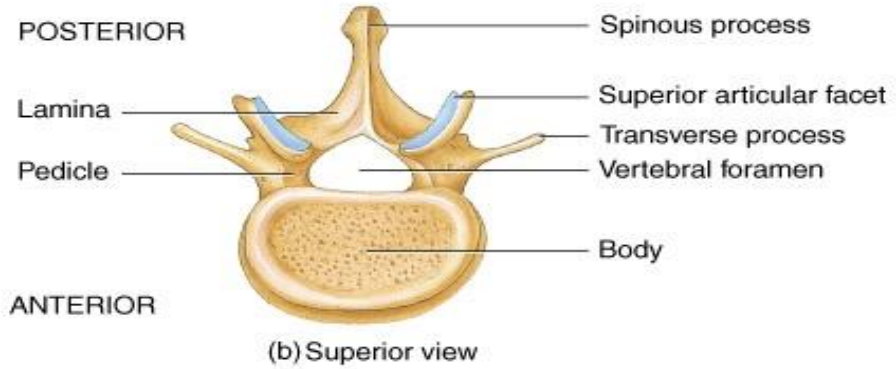
Dr Yasser Al Harbi

Learning Objectives

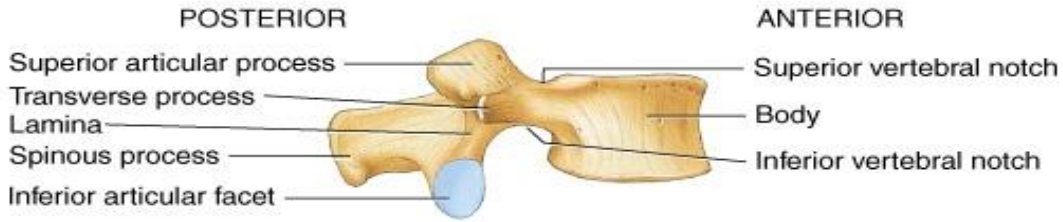
- Describe the structure of the low back: lumbar vertebrae, intervertebral discs, zygapophyseal joints, ligaments, vertebral canal, intervertebral foramen
- Describe the relationship of spinal cord and nerve roots to lumbar vertebral column
- Describe the innervation of the back and pain sensitive structures

Lumbar Vertebrae (L1-L5)

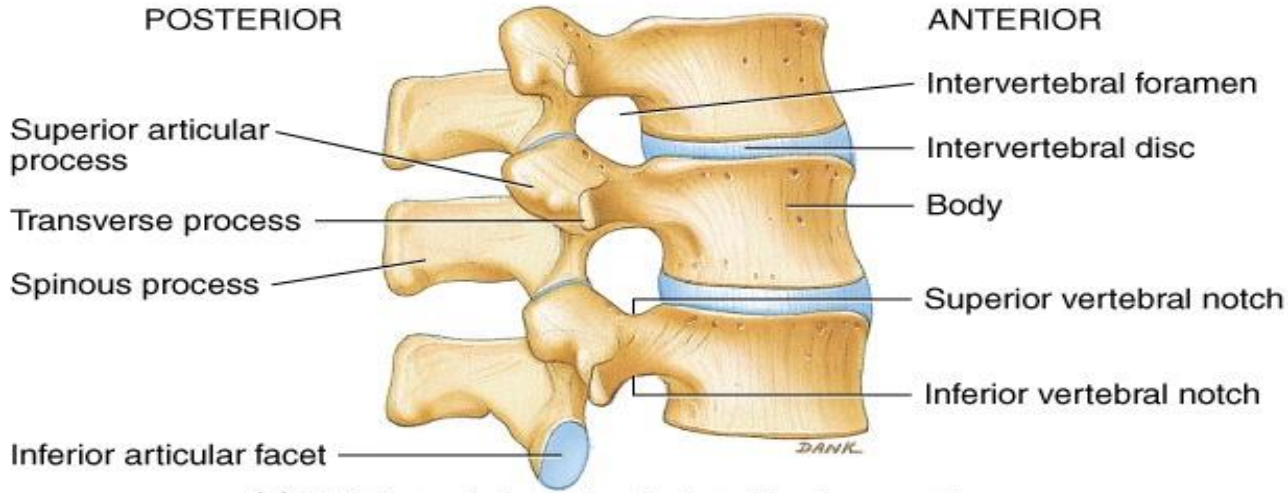
- Strongest & largest
- Body is large (kidney shape)
  - L5 is the largest
- Vertebral foramen is triangular in shape
  - cervical > lumbar > thoracic
- Transverse processes long and slender
  - Accessory process
    - Attachment of lumbar intertransversarii mm.
- Short thick spinous process (backward)
  - back musculature



(b) Superior view

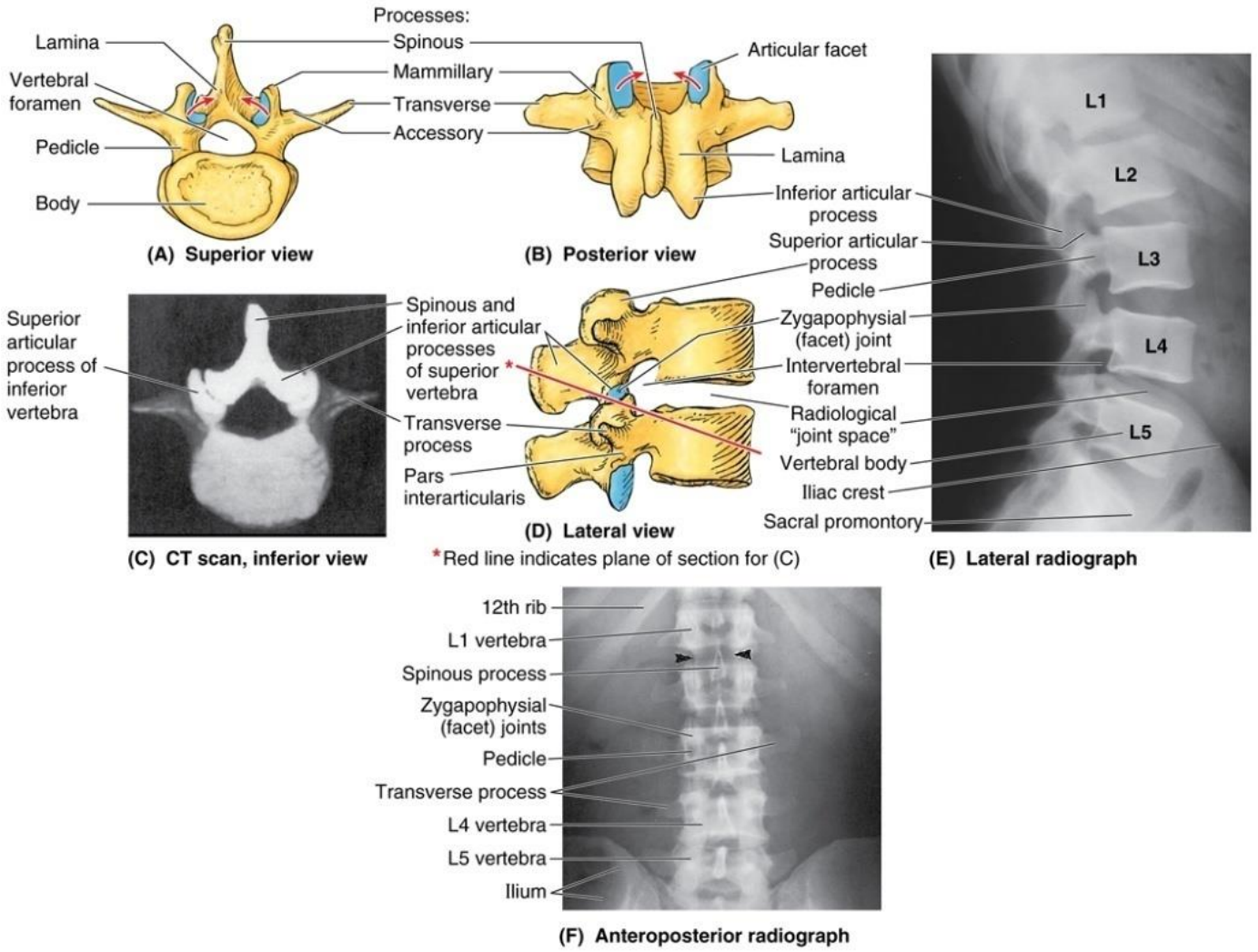


(c) Right lateral view



Lumbar Zygapophysial Joints

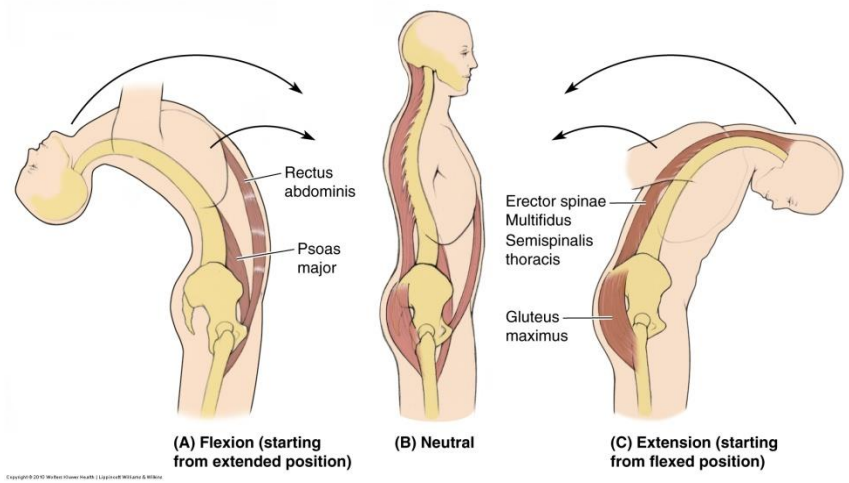
- Superior articular processes face medially
- Rotate medially with successive vertebrae until it faces posteriorly
- T12-L1 Zygapophysial Joints are sagittally oriented
- L5-S1 Zygapophysial Joints are coronally oriented

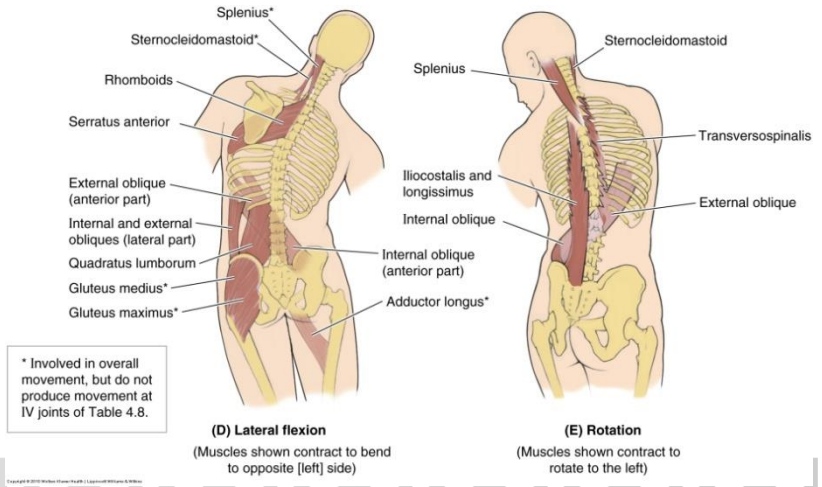


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Movements of Lumbar Vertebral Column

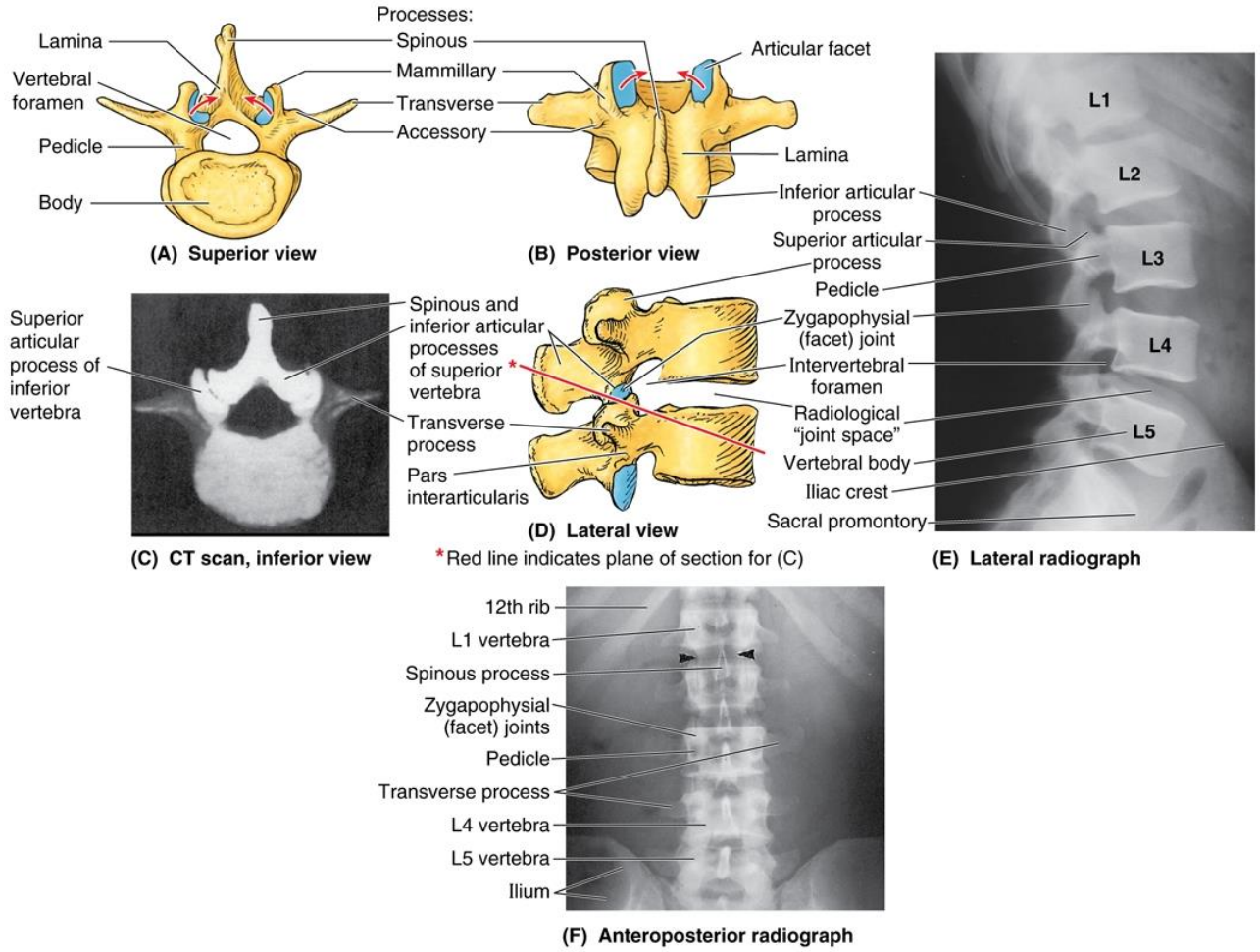
- Flexion – rectus abdominis m. & psoas major m.
- Extension – postvertebral mm.
- Lateral flexion – postvertebral mm., quaderatus lumborum m. & oblique abdominal mm.
- Rotation – rotators & oblique abdominal mm.





Lumbar IVF

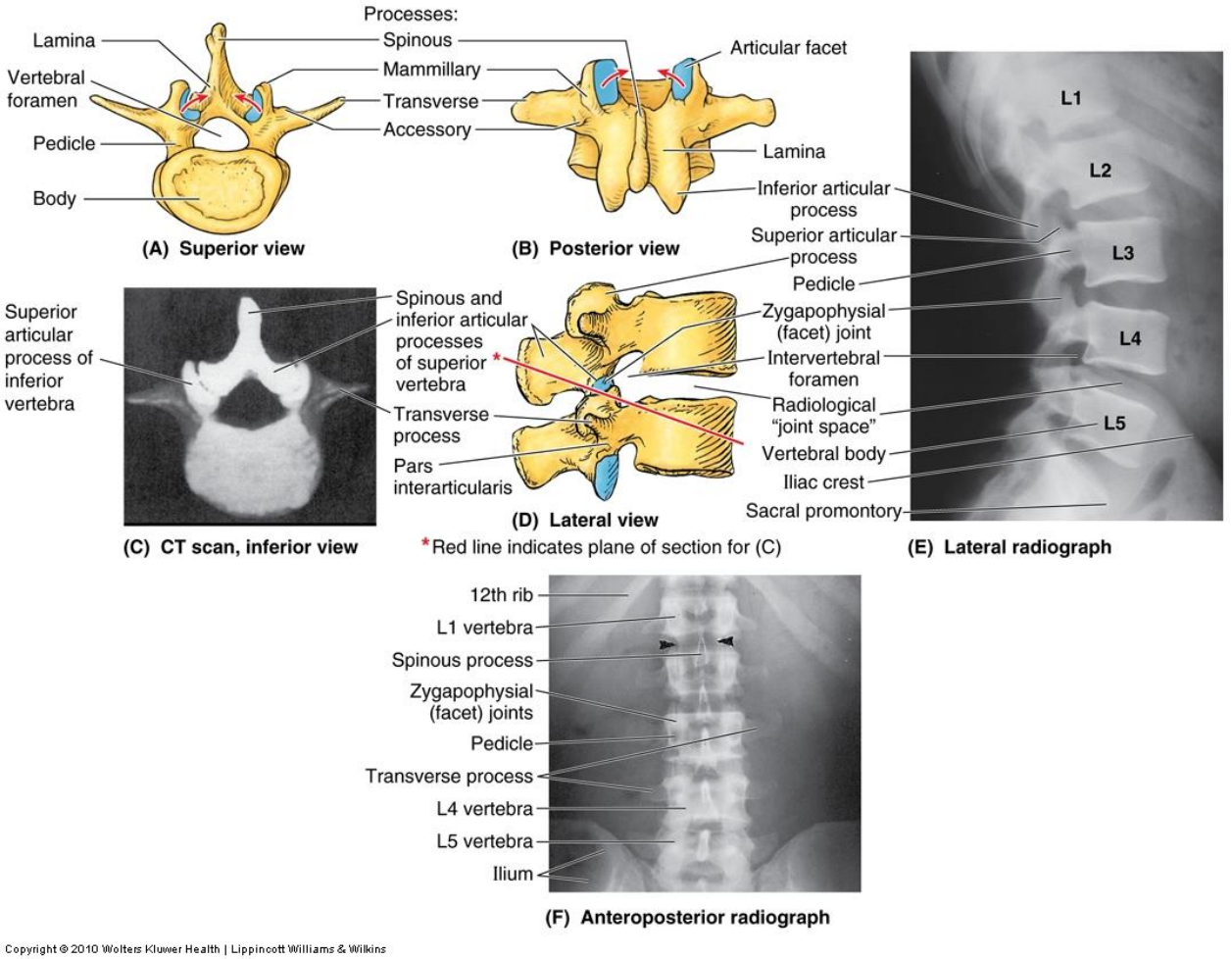
- IVF become narrower from L1 to L5
- Spinal nerves become larger from L1 to L5
- Increase the chance of nerve encroachment in the lower vertebrae due to IVF stenosis



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Lumbar IVD

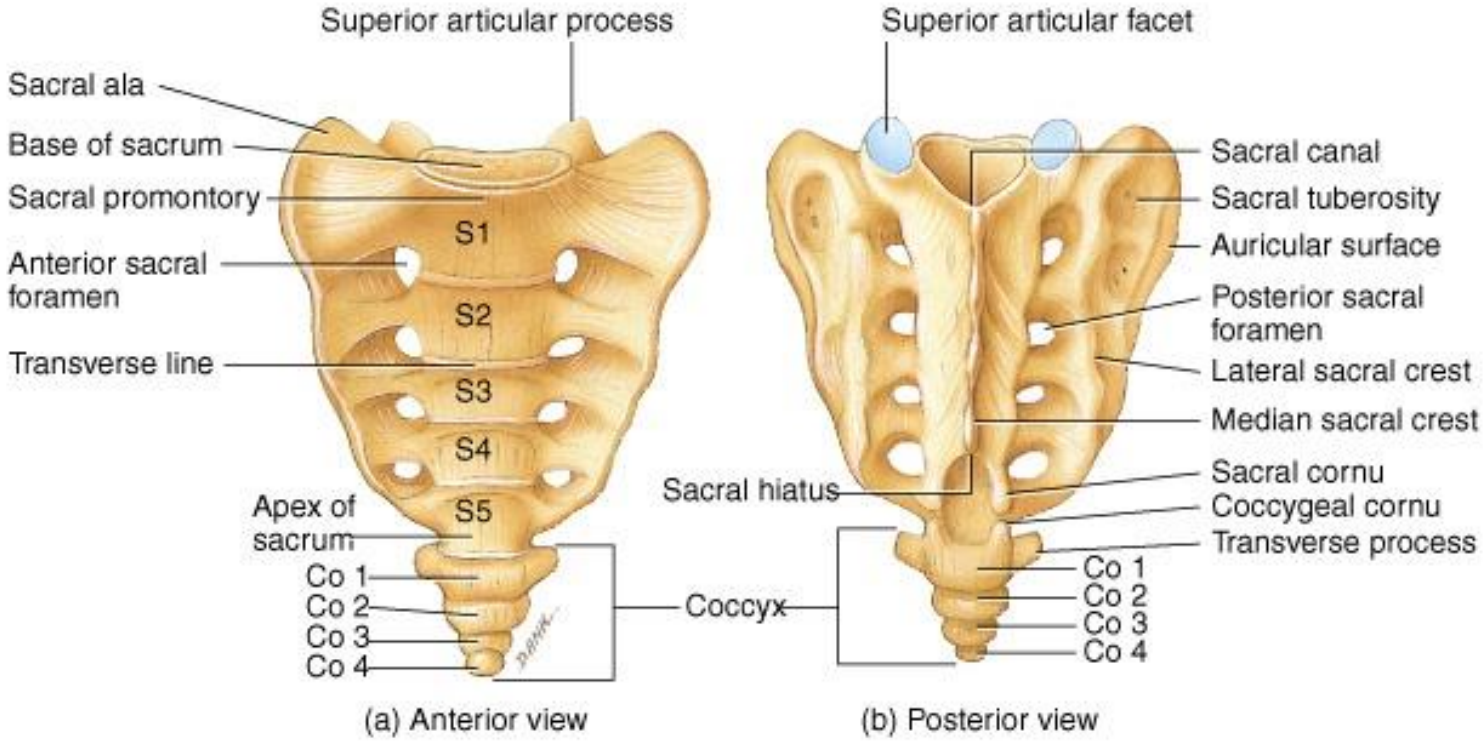
- Lumbar IVD are thicker compared with other regions
- Relative thickness of lumbar and cervical IVDs to the vertebral bodies are higher than thoracic ones
- Lumbar IVDs are thicker anteriorly to cope with the curvature
- Nucleus pulposus is more posteriorly located from the center
- Posterior part of the annulus fibrosus is weaker
- Herniation of IVD is mostly posterolateral
- Mostly affected IVD between L4-L5 or L5-S1

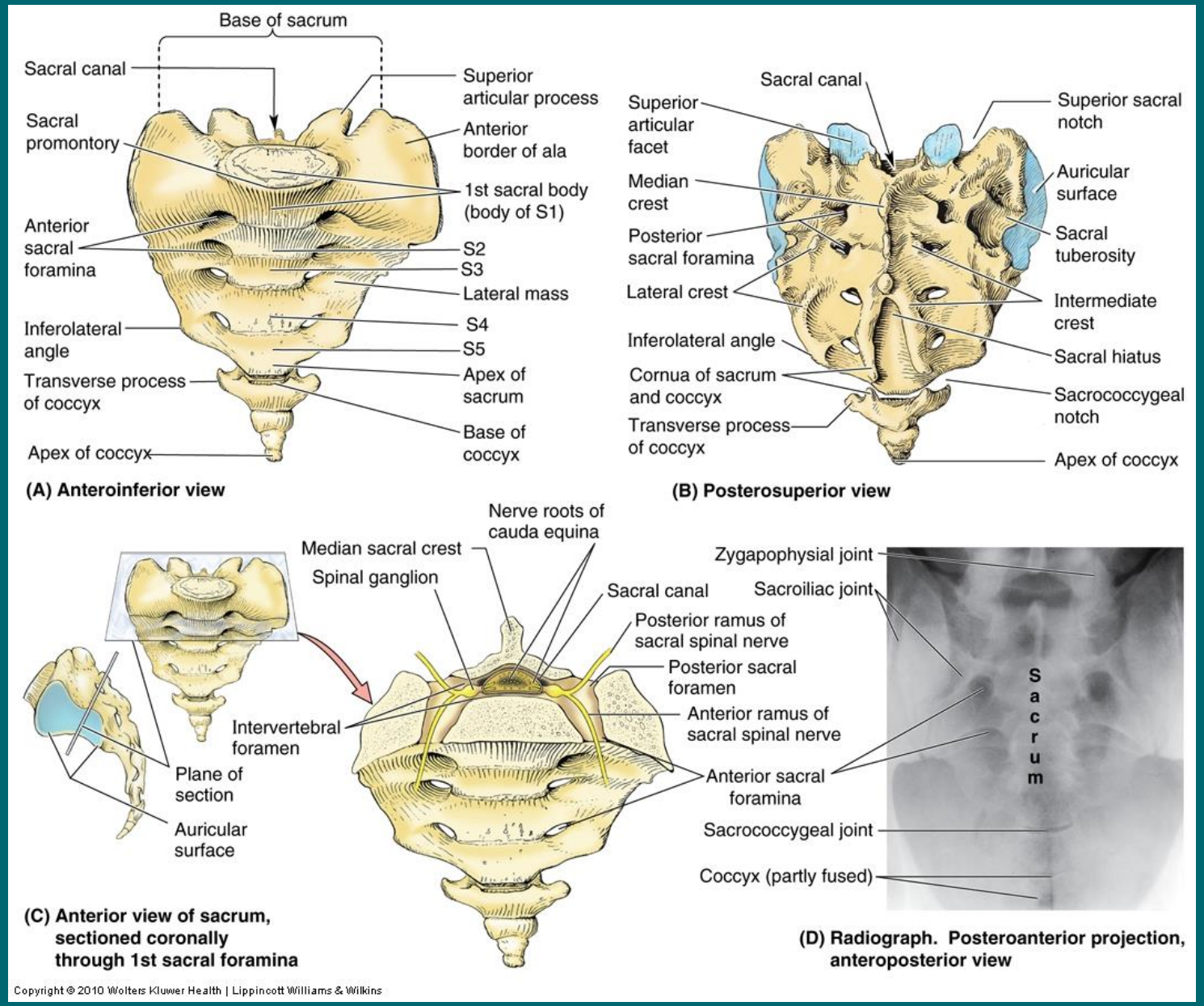


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Sacrum

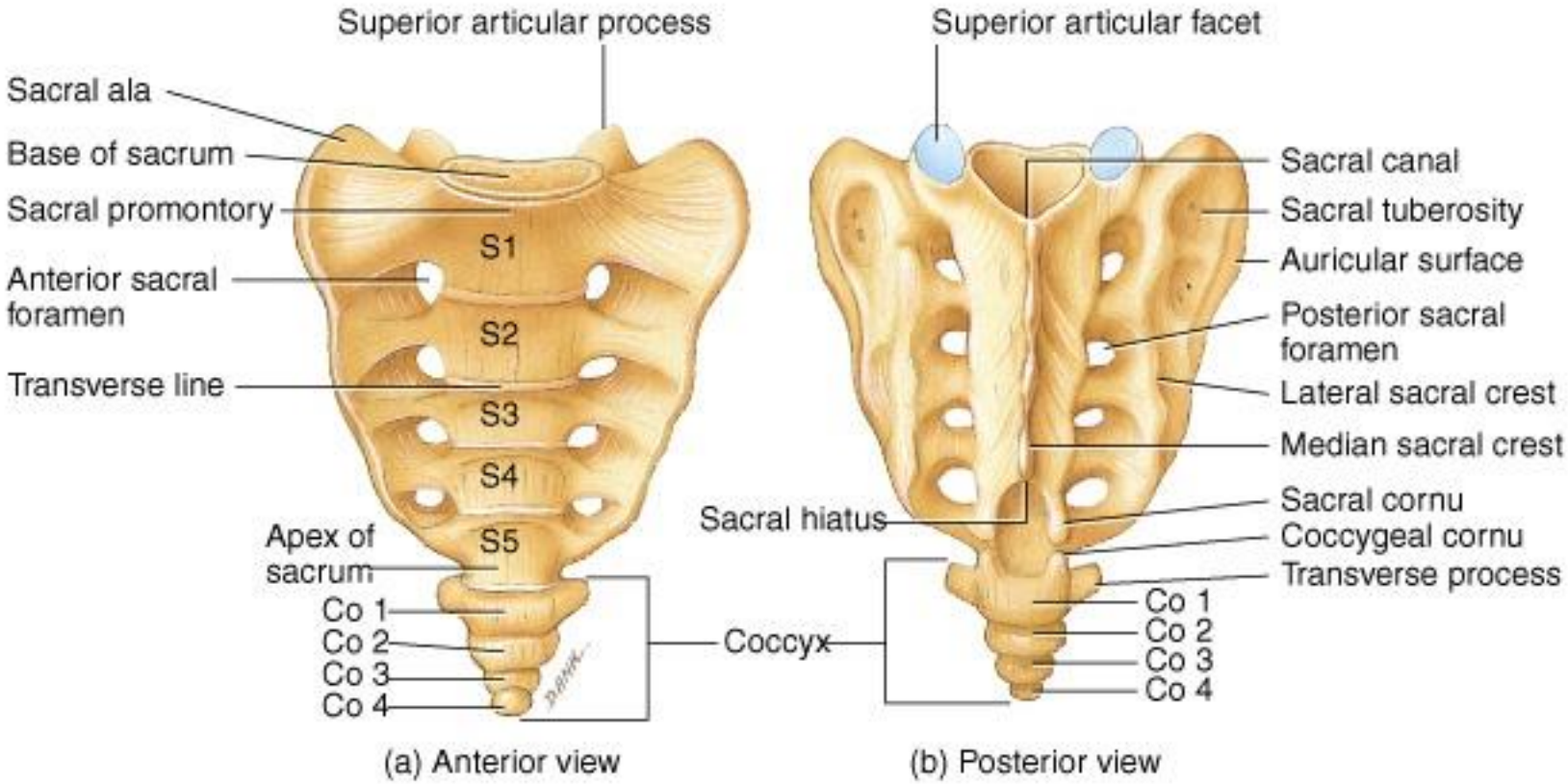
- Union of 5 vertebrae (S1 - S5) by age 30
  - median sacral crest was spinous processes
  - sacral ala is fused transverse processes
- Sacral promontory (anterosuperior margin)
- Sacral canal ends at sacral hiatus (laminae don't meet)
- Auricular surface & sacral tuberosity of sacroiliac joint
- Anterior and posterior sacral foramina
  - Ventral and dorsal rami of spinal nerves respectively





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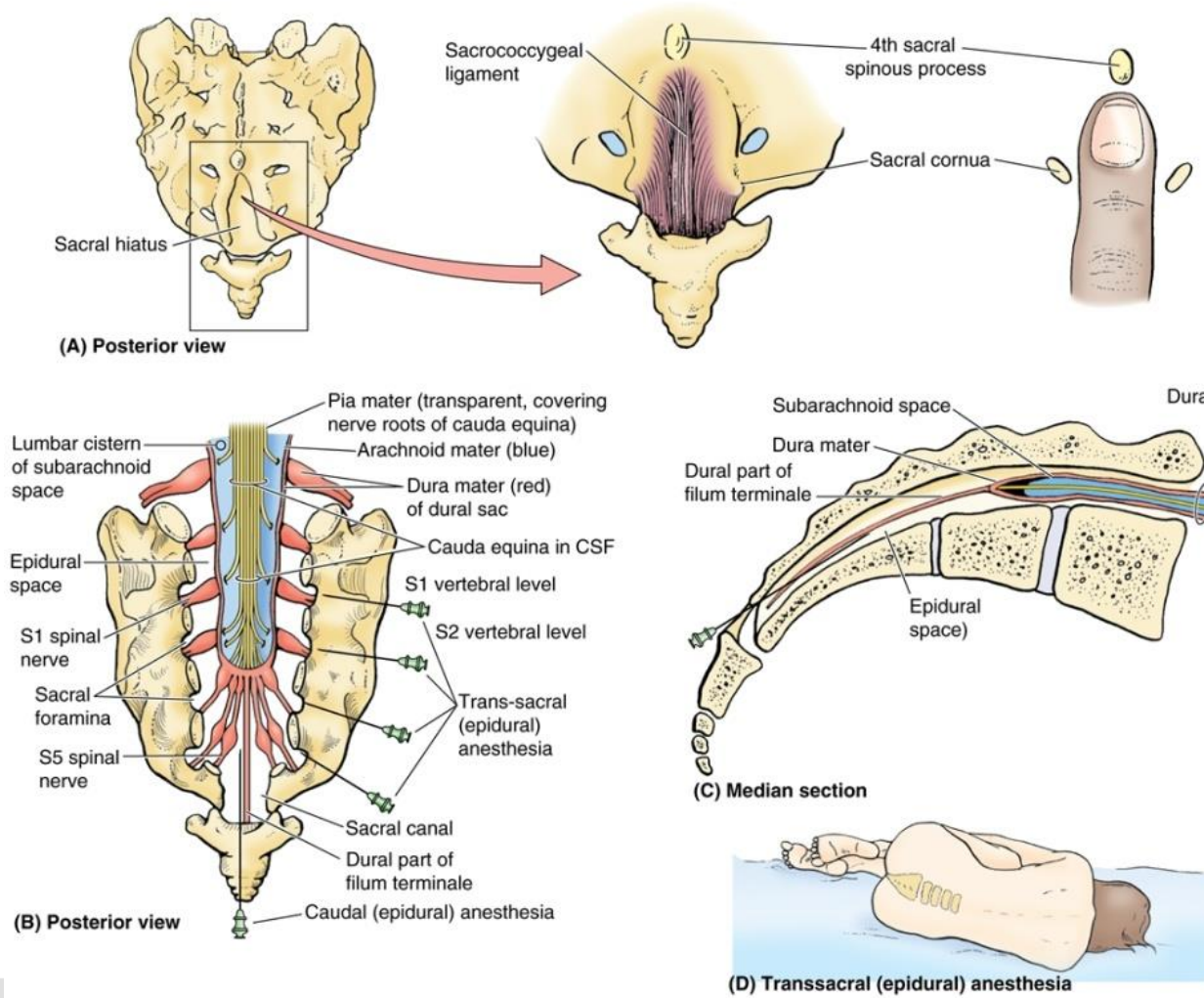
Coccyx



- Shape ..
- Coccygeal cornua
- Union of 4 vertebrae (Co1 - Co4) by age 30

Caudal Epidural Anesthesia

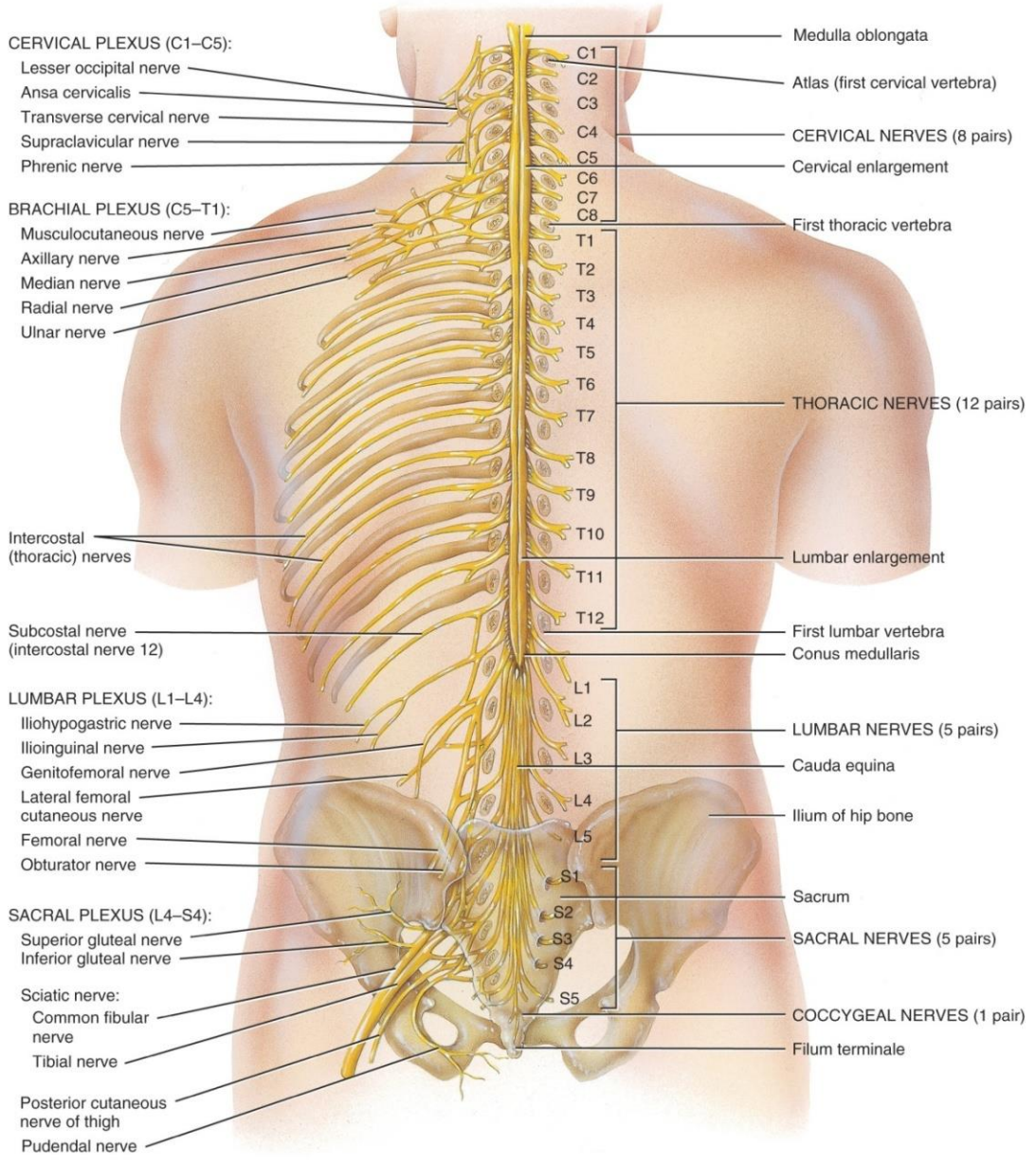
- Caudal epidural anesthesia during delivery
- Into sacral hiatus
- Sacral and coccygeal cornua are important landmarks
- Anesthetize S2-Co1 spinal nerves



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Inferior End of vertebral canal: Content

- Conus medullaris
  - In adult ends at L2
  - In newborn ends at L4
- Cauda equina (horse's tail)
  - dorsal & ventral roots of lowest spinal nerves (L1-Co1)
- Spinal meninges
  - Dura matter
    - Ends at S2-S3
  - Arachnoid matter
    - Ends at S2-S3
  - Subarachnoid space = CSF ➢
    - Lumbar cistern (enlargement in subarachnoid space) ➢
      - L2-S2 ➢
  - Pia matter
    - Filum terminale
      - thread-like extension of pia mater
      - stabilizes spinal cord in canal

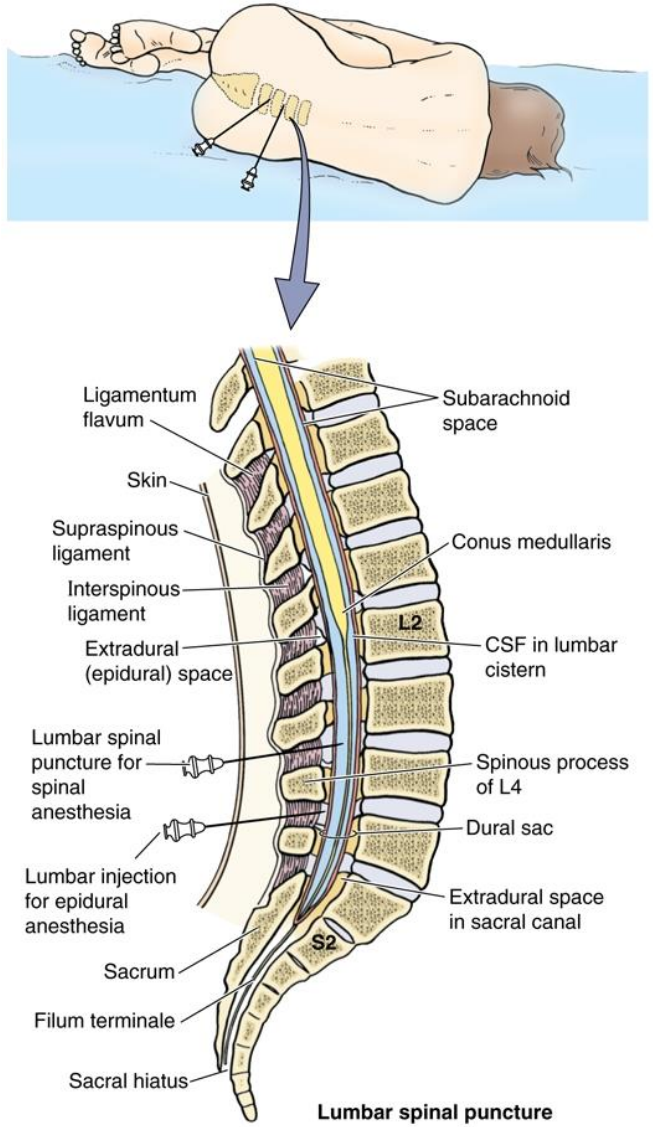


(a) Posterior view of entire spinal cord and portions of spinal nerves

Lumbar Puncture

- Lumbar puncture is used to withdraw CSF for diagnostic purposes
- LP performed from lumbar cistern to avoid the damage to the spinal cord
- LP approached mostly in L3-L4 or L4-L5

- Epidural anesthesia
- Target the epidural space
- Same approach as LP
- Could be approached from the sacral hiatus



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Spinal Nerves: Level of Exit

- From T1 to L5, spinal nerves exit from the IVF below their encountered vertebrae
- S1-S4 rami exit from their encountered sacral foramens
- S5 & Co1 exit from sacral hiatus

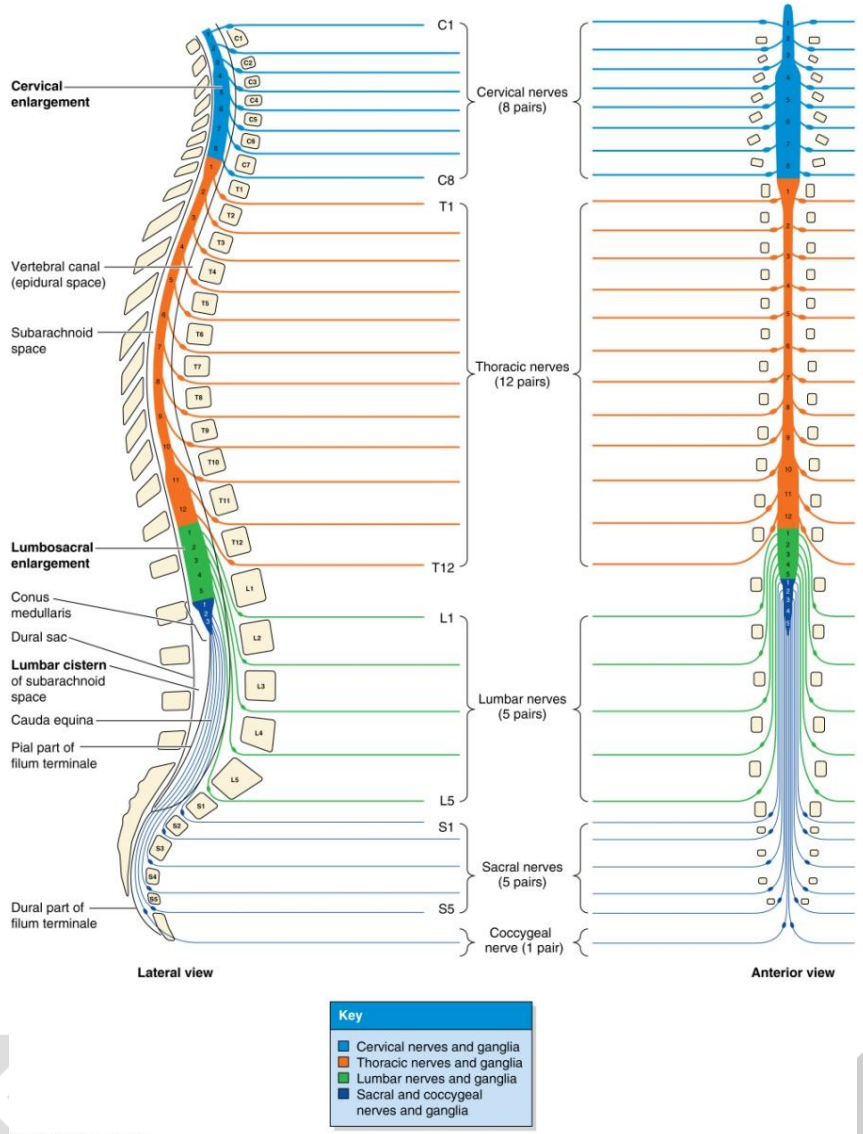


TABLE 4.13. NUMBERING OF SPINAL NERVES AND VERTEBRAE

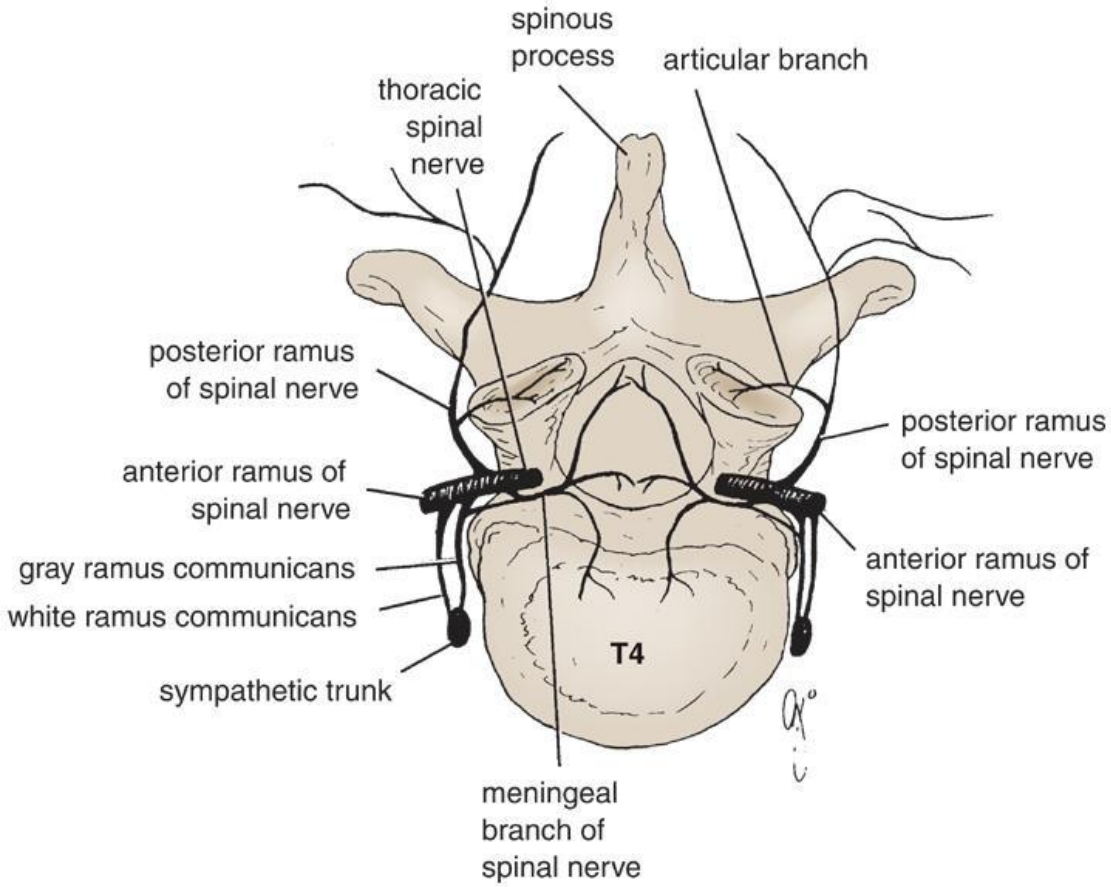
Segmental Level	Number of Nerves	Level of Exit from Vertebral Column
Cervical	8 (C1–C8)	Nerve C1[a] (suboccipital nerve) passes superior to arch of vertebra C1 Nerves C2–C7 pass through IV foramina superior to the corresponding vertebrae Nerve C8 passes through the IV foramen between vertebra C7 and T1
Thoracic	12 (T1–T12)	Nerves T1–L5 pass through IV foramina inferior to the corresponding vertebrae
Lumbar	5 (L1–L5)	
Sacral	5 (S1–S5)	Nerves S1–S4 branch into anterior and posterior rami within the sacrum, with the respective rami passing through the anterior and posterior sacral foramina
Coccygeal[a]	1 (Co1)	The 5th sacral and coccygeal nerves pass through the sacral hiatus

[a]The first cervical nerves lack posterior roots in 50% of people, and the coccygeal nerves may be absent.

(Modified from *Barr's The Human Nervous System.*)

Innervation of the Back

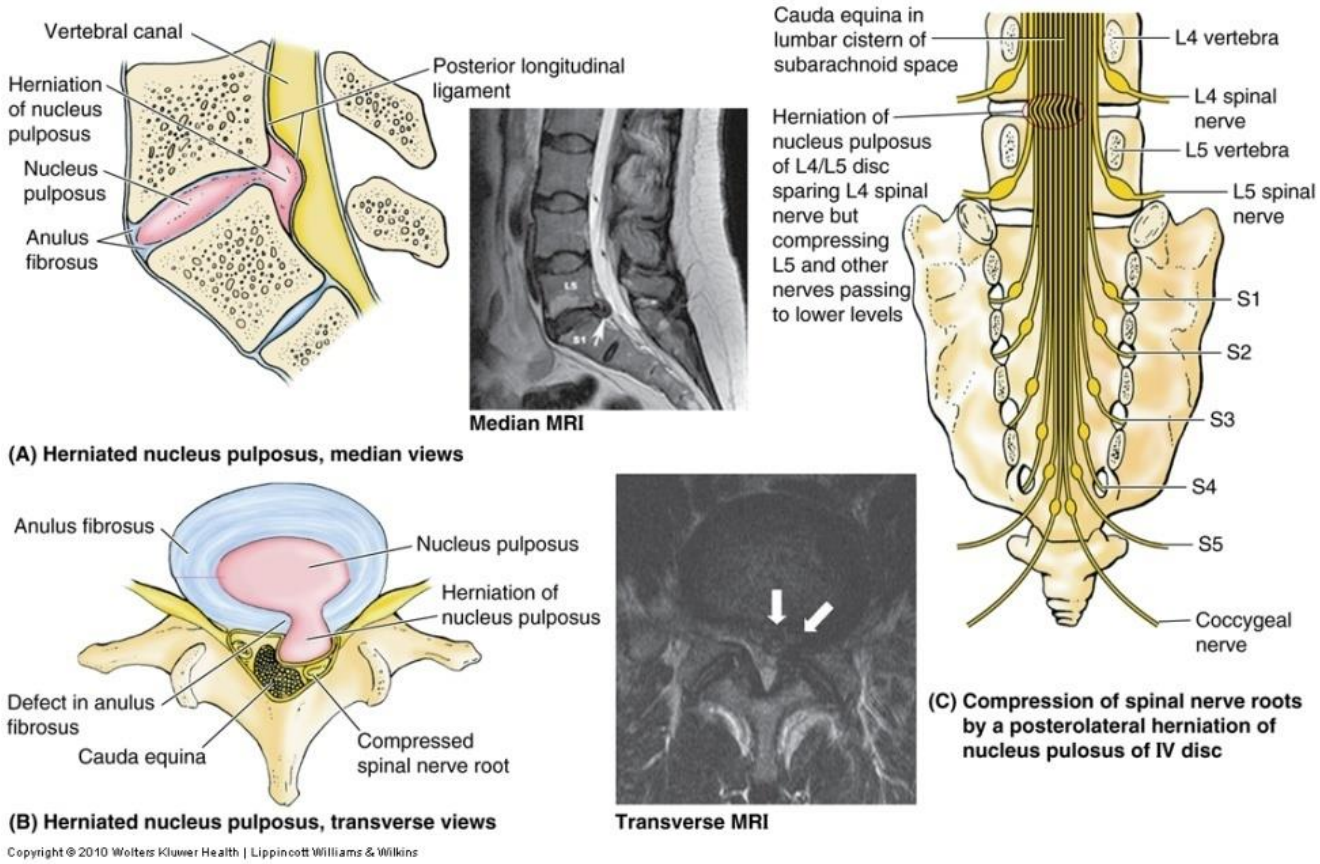
- Posterior rami of the spinal nerves
  - Cutaneous branches innervate the skin of the back
  - Muscular branches innervate the back muscles
- Dura matter innervated by the recurrent meningeal branches of the spinal nerves
  - Afferent and sympathetic fibers
- IVD is innervated by the meningeal branches
- Zygapophysial joints is innervated by the articular branches of the posterior rami



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Low Back: Pain Sensitive Structures

- Ligaments (posterior longitudinal ligament via desk herniation)
  - Acute low back pain
- Vertebral joints (due to inflammatory or degenerative process)
- Spinal nerves and their roots (due to disk herniation or IVF stenosis)
  - Sciatica- low back pain radiating to the posterior part of the thigh and the leg
- Back muscles (muscle strain) can cause severe acute non radiating low back pain



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Thanks